Respiratory Infections — Pneumonia, Influenza and COPD

1. Pneumonia — Overview

Pneumonia is an infection of the lung parenchyma causing consolidation (filling of alveoli with inflammatory exudate). It is the single largest infectious cause of death in children globally, accounting for 14% of all deaths in children under 5. In adults, it is a leading cause of hospitalisation and death.

Classification: Community-Acquired Pneumonia (CAP) — acquired outside hospital or within 48 hours of admission. Hospital-Acquired Pneumonia (HAP) — ≥48 hours after admission. Ventilator-Associated Pneumonia (VAP) — in intubated patients. Aspiration pneumonia — inhalation of oropharyngeal or gastric contents.

Common pathogens in CAP: Streptococcus pneumoniae (most common bacterial cause), Haemophilus influenzae, Staphylococcus aureus, Klebsiella pneumoniae (significant in India — associated with alcohol, diabetes), atypicals (Mycoplasma, Chlamydophila, Legionella — present more indolently).

2. Pneumonia — Symptoms and Severity Assessment

Classic symptoms: fever, cough (productive of yellow/green or rust-coloured sputum), pleuritic chest pain (sharp, worse with breathing and coughing), dyspnoea, tachypnoea. Non-specific: fatigue, myalgia, headache, diarrhoea, confusion (elderly).

Severity scoring: CURB-65 (Community Use): Confusion (new), Urea >7 mmol/L, Respiratory rate ≥30/min, Blood pressure <90/60 mmHg, Age ≥65. Score 0-1: low severity (outpatient), 2: moderate (consider hospital), 3-5: high severity (ICU consideration).

Children: WHO IMNCI criteria — fast breathing (>60/min <2 months, >50/min 2-12 months, >40/min 1-5 years), chest indrawing, grunting as severity markers. Very severe: central cyanosis, not able to drink, convulsions, very lethargic.

3. Pneumonia — Treatment

Mild-moderate CAP (outpatient): amoxicillin 1g TDS OR amoxicillin-clavulanate 625 mg TDS for 5 days. Atypical coverage: add azithromycin 500 mg OD x 5 days or doxycycline 100 mg BD. Respiratory fluoroquinolone (levofloxacin, moxifloxacin) as monotherapy for atypical or penicillin-allergic.

Severe CAP (inpatient): IV beta-lactam (co-amoxiclav, ceftriaxone 1-2g) + IV/oral azithromycin or doxycycline. ICU: IV beta-lactam + IV azithromycin OR IV beta-lactam + IV fluoroquinolone.

Children: amoxicillin oral for non-severe pneumonia (preferred over cotrimoxazole in revised WHO guidelines). IV benzylpenicillin or ampicillin for severe. Gentamicin + ampicillin for very severe or neonatal pneumonia.

Supportive: oxygen (target SpO_2 >94%, ≥90% in COPD), adequate hydration, antipyretics. Physiotherapy for sputum retention. Vaccination: pneumococcal conjugate vaccine (PCV13/PCV15) and influenza vaccine prevent a significant proportion of pneumonia cases.

4. Influenza

Influenza (flu) is caused by influenza A and B RNA viruses. Influenza A (H1N1, H3N2 subtypes) and B circulate seasonally. India sees monsoon and post-monsoon peaks. Pandemic influenza — global outbreaks due to antigenic shift (2009 H1N1 'Swine Flu' pandemic).

Clinical features: abrupt onset of high fever, severe myalgia, malaise, headache, non-productive cough, sore throat, rhinorrhoea. Can cause primary viral pneumonia or secondary bacterial pneumonia. High-risk groups for complications: age >65, <5 years, pregnancy, immunocompromised, chronic cardiopulmonary disease.

Treatment: oseltamivir (Tamiflu) 75 mg BD x 5 days — most effective if started within 48 hours of symptoms. High-risk patients and all hospitalised cases should receive antiviral treatment regardless of timing.

Prevention: annual influenza vaccine (strongly recommended for all health workers, high-risk groups, pregnant women). Vaccines are reformulated annually to match predicted circulating strains.

5. COPD — Chronic Obstructive Pulmonary Disease

COPD is a preventable and treatable lung disease characterised by persistent, usually progressive airflow limitation due to airway and/or alveolar abnormalities from significant exposure to noxious particles or gases.

In India, COPD affects approximately 15 million people. Uniquely, India has substantial COPD due to indoor air pollution from biomass fuel burning (wood, dung, crop residue) — affecting mostly non-smoking rural women. Tobacco smoking remains the leading cause globally.

Pathophysiology: chronic bronchitis — mucus hypersecretion and chronic inflammation of bronchi. Emphysema — destruction of alveolar walls, loss of elastic recoil, air trapping.

Spirometry: the gold standard for COPD diagnosis. Post-bronchodilator FEV1/FVC <0.70 confirms persistent airflow limitation. GOLD grades (1-4) based on FEV1% predicted. GOLD groups (A-D) incorporate symptoms (CAT score, mMRC dyspnoea scale) and exacerbation history.

6. COPD — Treatment

COPD Group	Recommended Initial Therapy
Group A (low risk, low symptoms)	Short-acting bronchodilator (SABA or SAMA) as needed
Group B (low risk, more symptoms)	Long-acting bronchodilator (LAMA or LABA)
Group E (high risk)	LAMA + LABA; consider adding ICS if eos ≥300/µL or ≥300 eos with exacerbations

Long-acting bronchodilators: LAMAs (tiotropium, umeclidinium, glycopyrronium) and LABAs (indacaterol, olodaterol, salmeterol, formoterol). LAMA is preferred over LABA. Combination LAMA+LABA provides additive bronchodilation.

Inhaled corticosteroids (ICS): do NOT use ICS alone in COPD. Add ICS to LAMA+LABA in patients with frequent exacerbations and eosinophils ≥300/µL (or ≥100/µL with frequent exacerbations). Risk of pneumonia with ICS — use only when indicated.

Non-pharmacological: smoking cessation (most important intervention — slows FEV1 decline). Pulmonary rehabilitation (exercise training, education, self-management — improves exercise capacity and quality of life). Long-term oxygen therapy (LTOT): for patients with PaO2 ≤55 mmHg at rest (>15 hours/day).

7. COPD Exacerbation Management

Acute exacerbations (AECOPD): acute worsening of respiratory symptoms beyond normal day-to-day variation. Triggers: respiratory viral infections (most common), bacterial infections, air pollution. Classified: mild (managed at home), moderate (hospitalisation), severe (ICU).

Outpatient management: increase SABA (salbutamol/ipratropium nebulisation). Oral prednisolone 40 mg/day x 5 days (not 14 days — equivalent benefit, fewer side effects). Antibiotics (amoxicillin, doxycycline, azithromycin) if purulent sputum or increased dyspnoea and sputum.

Hospital management: controlled oxygen therapy (target SpO_2 88-92% — avoid high-flow oxygen which can suppress hypoxic ventilatory drive in chronic CO_2 retainers). Non-invasive ventilation (NIV — BiPAP) for hypercapnic respiratory failure (pH <7.35 + $PaCO_2$ >45 mmHg). IV aminophylline if no response to bronchodilators.

8. Indoor Air Pollution and COPD in India

Indoor air pollution from biomass burning (cooking fires) is responsible for approximately 17% of COPD in India. Over 600 million people in India still rely on solid fuels for cooking. Concentration of $PM_{2.5}$ from cooking fires can exceed 1000 µg/m^3 — far above the WHO safe limit of 15 µg/m^3.

The Pradhan Mantri Ujjwala Yojana (PMUY) distributes LPG connections to below-poverty-line households. This initiative has provided over 90 million LPG connections. Studies show reduction in indoor pollution and respiratory disease with LPG transition.

COPD is also significantly driven by outdoor air pollution in Indian cities. AQI (Air Quality Index) monitoring, vehicle emission standards (BS-VI standards now in effect), industrial controls, and urban planning for green spaces are important public health interventions.

9. Acute Respiratory Distress Syndrome (ARDS)

ARDS is a severe form of acute diffuse lung injury leading to non-cardiogenic pulmonary oedema, bilateral infiltrates, severe hypoxemia (PaO_2/FiO_2 ratio <300 — mild; <200 — moderate; <100 — severe), and respiratory failure.

Common triggers: sepsis (most common), pneumonia (including COVID-19), aspiration, pancreatitis, trauma, blood transfusion (TRALI), burns.

Management (in ICU): lung-protective mechanical ventilation — low tidal volume (6 ml/kg IBW), plateau pressure <30 cm H_2O, PEEP titrated to oxygenation and compliance. Prone positioning for 12-16 hours/day in moderate-severe ARDS improves mortality (PROSEVA trial). Conservative fluid strategy after initial resuscitation. No proven pharmacotherapy except dexamethasone in COVID-ARDS.

10. Respiratory Health in India — Public Health Perspective

India has among the world's highest levels of outdoor air pollution — 21 of the world's 30 most polluted cities are in India (IQ Air 2023). Respiratory diseases including COPD, asthma, lung cancer, and bronchiectasis are direct consequences.

National Clean Air Programme (NCAP): launched 2019, aims for 20-30% reduction in PM2.5 and PM10 by 2024. Focuses on 132 non-attainment cities. Challenges include industrial emissions, vehicle exhaust, crop burning (stubble burning in Punjab-Haryana causing Delhi's severe November smog), construction dust.

Tobacco control: Cigarettes and Other Tobacco Products Act (COTPA) 2003 restricts advertising, sales to minors, and requires pictorial health warnings (85% pack coverage). India's smoking rate has declined but smokeless tobacco (gutka, khaini, betel quid) use remains very high.

Universal vaccination: Pentavalent vaccine (DTP-HB-Hib) and PCV (Pneumococcal Conjugate Vaccine) are part of India's Universal Immunisation Programme and reduce bacterial respiratory infections significantly in children.